

Quantum Computing (CS-4084)

Date: 3rd June, 2026

Course Instructor(s)

Dr. Faisal Aslam

Final Examination

Total Time: 3 Hours

Total Marks: 54

Total Questions: 6

22L-6790

Roll No

8A

Section

Signature

Instructions: One A4 handwritten cheatsheet allowed. Write roll number clearly.

Attempt all questions.

CLO 1: Understand fundamental principles of quantum information and quantum mechanics.

1. Shor 9-Qubit Code**[8 marks]**

The Shor code protects against bit-flip and phase-flip errors by concatenation.

Suppose a student reverses the order: first phase-flip protection, then bit-flip protection.

(a) Describe or draw the encoding process.

[4 marks]

(b) Will the resulting 9-qubit code still be able to correct any single-qubit Pauli error (X , Y , or Z)? Explain briefly.

[4 marks]

CLO 2: Learn about quantum gates and circuits.

2. Syndrome Measurement**[6 marks]**

3-qubit bit-flip code stabilizers: $S_1 = Z_1 Z_2$, $S_2 = Z_2 Z_3$.

(a) Give the syndrome (as ± 1 pairs) for errors X_1 , X_2 , X_3 .

[4 marks]

(b) Can these uniquely locate a bit-flip? Justify.

[2 marks]

CLO 3: Develop a deep comprehension of quantum computation and its associated algorithms.

3. Order Finding (Shor's Algorithm)**[8 marks]**

$f(x) = 2^x \bmod 15$.

(a) Compute $f(0)$, $f(1)$, $f(2)$, $f(3)$, $f(4)$, $f(5)$.

[1 mark]

(b) Find the order r of 2 modulo 15.

[3 marks]

(c) Use r to factor 15. Show your working.

[4 marks]

CLO 3: Develop a deep comprehension of quantum computation and its associated algorithms.

4. Period Finding**[8 marks]**

Assume the first register contains 4 qubits. After measuring the second register, the first register collapses to

$$\frac{1}{2} (|1\rangle + |5\rangle + |9\rangle + |13\rangle).$$

(a) What is the period?

[2 marks]

(b) Why do only these states remain after measurement?

[3 marks]

(c) What information does the QFT extract from this state?

[3 marks]

CLO 3: Develop a deep comprehension of quantum computation and its associated algorithms.

5. Grover's Algorithm

[8 marks]

$N = 16$, one marked item.

(a) Compute $\theta = \sin^{-1}(1/\sqrt{N})$.

[2 marks]

(b) By how much does one Grover iteration increase the state's angle toward the marked state? [2 marks]

(c) Approximately how many Grover iterations should be applied?

[4 marks]

CLO 3: Develop a deep comprehension of quantum computation and its associated algorithms.

6. Short Questions

[16 marks]

(2 marks each).

- (a) Why is measurement in phase estimation usually delayed until after the QFT?
- (b) Why does the no-cloning theorem prevent a simple backup strategy for quantum computers?
- (c) Why is an odd order usually discarded in Shor's algorithm?
- (d) Why is the QFT useful for period finding but not directly useful for unstructured search?
- (e) Why must phase estimation begin with an eigenstate (or contain an eigenstate component)?
- (f) Why does Grover's algorithm provide quadratic rather than exponential speedup?
- (g) Why are both quantum and classical computations required in Shor's algorithm?
- (h) Why does syndrome measurement not reveal the encoded logical quantum state?